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Joint detection, tracking and communication optimization based on radar-communication UAV

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ABSTRACT

Improving the ability of an unmanned aerial vehicle (UAV) to perform tasks is of great significance in the future battlefield. This paper investigates the detection, tracking, and communication tasks performed by a radar-communication UAV under time and energy constraints. To balance the constraints and task performance, we formulate a joint detection, tracking, and communication optimization problem in a UAV-assisted radar and communication (RC) system by jointly optimizing dwell time, scheduling interval, and UAV trajectory. Due to the non-convex problem, we propose an alternative iterative optimization (AIO) algorithm to find the suboptimal solution. Simulation results demonstrate that the proposed algorithm can obtain higher joint performance than baseline schemes while reaching convergence.

1. Introduction

Nowadays, an unmanned aerial vehicle (UAV) is becoming popular to play an important role in future battlefield networks that can provide both ubiquitous communication [1] and radar sensing functions [2]. On the battlefield, joint UAVs with different functions are needed to perform complex tasks to obtain enemy intelligence. However, the combination of different functional UAVs reduces work efficiency and increases resource consumption. Therefore, multi-functional UAVs could be a research direction to solve this problem, for instance, the radar-communication UAV [3].

Considering the advantages of radar-communication UAVs, some scholars have started to apply them in several areas, e.g., civilian and military applications. In [4], the authors studied the dual-functional multi-UAV network utility maximization problem, which was solved by a computationally practical method. The authors in [5] proposed a novel integrated scheduling method for mmWave/THz communications in UAV networks to enable data transmission from UAV to the ground base station.

The literature above introduces the application of radar-communication UAVs at the auxiliary rather than the task level. For the task level, the authors in [6] studied the joint task allocation and resource optimization problem in the multi-UAV system solved by a loop iterative optimization (LIO) algorithm to obtain the suboptimal solution, i.e., UAVs are rationally assigned to perform the detection, tracking, and communication tasks. However, [6] does not consider the radar-communication UAV for task execution. Based on [6,7] introduced radar-communication UAVs and a task web to study the joint task allocation and resource optimization problem, also aiming to obtain a suboptimal solution for assigning appropriate task chains to perform detection, tracking, and communication tasks. Overall, although the literature [6,7] has focused on improving task performance, they have not considered optimizing the task execution time and time interval during task execution.

In this paper, we study a UAV-assisted radar and communication (RC) system where a radar-communication UAV is used to perform the detection, tracking, and communication tasks in a specific order during a given finite period. The objective here is to maximize the joint performance among these tasks under time and energy constraints. To this extent, we consider a joint detection, tracking, and communication optimization problem by jointly optimizing dwell time, scheduling interval, and UAV trajectory. To this end, we propose an alternative iterative optimization (AIO) algorithm with polynomial time complexity to find the suboptimal solution. The proposed algorithm will be verified by simulation results.

2. System model and problem formation

2.1. System model

We consider a UAV-assisted RC system which consists of a radar-communication UAV, a target, and a base station. In the system, the

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radar in the UAV acquires the target intelligence by detecting and tracking the target, while the communication equipment in the UAV transmits the acquired target information to the base station. To better understand the system, there are some assumptions: (i) the UAV and the target fly at $\mathbf{q}_{\mathrm{u}} = [x_1, y_1, H]^{\mathrm{T}}$ and $\mathbf{q}_{\mathrm{t}} = [x_2, y_2, H]^{\mathrm{T}}$, respectively; (ii) the velocity of the target is v, and the maximum velocity of the UAV is $v_{\max}$; (iii) the UAV is hovering during the detection and tracking tasks and flying during the communication task, respectively; (iv) the UAV and the target are moving in the opposite direction; (v) the base station is fixed at $\mathbf{Z}_0 = [x_0, y_0, 0]^{\mathrm{T}}$.

We assume that the total time and the scheduling interval are T and t_{SI}, respectively, where t_{SI} is defined as the time period to perform a task set. The task set is defined as $\mathcal{T} = \{$Detection,Tracking, Communication$\}$, where $M = 3$ denotes the size of the task set. N is the number of scheduling intervals. At the end of each scheduling interval, both the UAV and the target return to their initial positions. Assume that the dwell time of task i at a scheduling interval is t_i ($i \in \mathcal{M} \triangleq \{1, 2, \ldots, M\}$). It is noted that the switching time among tasks is ignored. In this letter, we only consider time and energy constraints.

2.2. Task model

2.2.1. Detection

Assume that the radar in the UAV transmits a single pulse for target detection at each scheduling interval. Under the Swerling I-type target, the detection probability of the UAV is

$$p_{\mathrm{d}} = \sqrt[(1+\mathrm{SNR})]{p_{\mathrm{fa}}} \tag{1}$$

where p_{fa} is the false alarm probability; SNR denotes the receiver echo signal-to-noise ratio, which is calculated by the radar equation and defined as

$$\mathrm{SNR} \propto \frac{1}{\bar{R}} \tag{2}$$

where the average distance between the UAV and the target is expressed as $\bar{R} = \frac{\|\mathbf{q}_{\mathrm{u}} - \mathbf{q}_{\mathrm{t}}\|_2 - vt_1}{2}$, and $\|\cdot\|_2$ is 2-Norm. Besides, the energy consumption of the detection task is $E_1 = P_{\mathrm{r}} t_1$, where P_{r} is the transmitter power when the system works in radar mode.

2.2.2. Tracking

Assume that the target is moving with uniform velocity, and the target motion state $\mathbf{x}$ is a four-dimensional vector, i.e., $\mathbf{x} = [x_2, y_2, \dot{x}_2, \dot{y}_2]^T$, where $\dot{x}_2$ and $\dot{y}_2$ are the velocity components of the x-axis and y-axis, respectively, and $\dot{x}_2 = v, \dot{y}_2 = 0$. When at m moment, the target motion state equation is [8]

$$\mathbf{x}(m+1) = \mathbf{F}\mathbf{x}(m) + \mathbf{G}y(m) \tag{3}$$

where

$$\mathbf{F} = \begin{bmatrix} 1 & T_0 \\ 0 & 1 \end{bmatrix}, \mathbf{G} = \begin{bmatrix} T_0^2/2 \\ T_0 \end{bmatrix} \tag{4}$$

and $y(m)$ is a Gaussian noise with a mean of 0 and a variance of σ_{v}^2; T_0 denotes the tracking sampling interval. Assume that ΔV and ΔX are the target velocity variable and the target position variable, respectively. Based on (3)~(4), we obtain $\Delta V \sim N(0, \sigma_{\mathrm{v}}^2 T_0^2)$ and $\Delta X \sim N(\dot{x}_2 T_0, \frac{1}{4}\sigma_{\mathrm{v}}^2 T_0^4)$, i.e., ΔV follows a Gaussian distribution with a mean of 0 and a variance of $\sigma_{\mathrm{v}}^2 T_0^2$ and ΔX follows a Gaussian distribution with a mean of $\dot{x}_2 T_0$ and a variance of $\frac{1}{4}\sigma_{\mathrm{v}}^2 T_0^4$.

According to the Gaussian distribution of the target velocity and the Gaussian distribution of the target position above, the velocity-association probability and the position-association probability of the UAV to perform the tracking task are [9]

$$p_{\mathrm{v}} = \int_{-L_{\mathrm{v}}}^{L_{\mathrm{v}}} \frac{1}{(2\pi)^{1/2}\sigma_{\mathrm{v}} T_0} \exp\left(-\frac{z^2}{2\sigma_{\mathrm{v}}^2 T_0^2}\right) dz \tag{5}$$

$$p_{\mathrm{x}} = \int_{-L_{\mathrm{x}}}^{L_{\mathrm{x}}} \frac{2}{(2\pi)^{1/2}\sigma_{\mathrm{v}} T_0^2} \exp\left(-\frac{2(z - vT_0)^2}{\sigma_{\mathrm{v}}^2 T_0^4}\right) dz \tag{6}$$

where L_{v} denotes radial velocity correlation threshold; L_x represents distance correlation threshold. Since there is only one tracking task in each scheduling interval, we have $T_0 = t_{\mathrm{SI}}$.

Referring to (5)~(6), the tracking performance of the UAV is defined as

$$\begin{aligned} p_g &= \frac{t_2 p_{\mathrm{v}} p_{\mathrm{x}}}{t_{\mathrm{SI}}} \\ &= \frac{t_2 \int_{-L_{\mathrm{v}}}^{L_{\mathrm{v}}} \exp\left(-\frac{z^2}{2\sigma_{\mathrm{v}}^2 t_{\mathrm{SI}}^2}\right) dz \int_{-L_{\mathrm{x}}}^{L_{\mathrm{x}}} \exp\left(-\frac{2(z - vt_{\mathrm{SI}})^2}{\sigma_{\mathrm{v}}^2 t_{\mathrm{SI}}^4}\right) dz}{\pi \sigma_{\mathrm{v}}^2 t_{\mathrm{SI}}^4} \end{aligned} \tag{7}$$

Moreover, the energy consumption of the tracking task is $E_2 = P_{\mathrm{r}} t_2$.

2.2.3. Communication

Assume that the communication time of the UAV in a scheduling interval is divided into N_1 equal short time slots, and the time slot is $\gamma = \frac{t_3}{N_1}$. Since the time slots are small enough, the UAV positions within each time slot are the same. In the nth time slot, the UAV position is expressed as $\mathbf{q}_{\mathrm{u}}(n) = [x_1(n), y_1(n), H]^{\mathrm{T}}$ ($n \in \{1, 2, \ldots, N_1\}$). Then, the channel gain between the UAV and base station follows the free-space path loss model, which can be written as

$$h(n) = \frac{\rho_0}{H^2 + \|\mathbf{q}_{\mathrm{u}}(n) - \mathbf{Z}_0\|_2^2} \tag{8}$$

where ρ_0 denotes the channel gain at reference distance ($d = 1$ m). Then, the transmission rate of the UAV at the nth time slot is expressed as

$$R_n = \gamma B \log_2\left(1 + \frac{P_{\mathrm{c}} h(n)}{n_0 B}\right) \tag{9}$$

where B and n_0 denote the channel bandwidth and the power spectral density of additive white Gaussian noise, respectively; P_{c} is the transmit power when the system works in communication mode.

Besides, the communication performance and the energy consumption of the communication task are $R_{\mathrm{c}} = \frac{\sum_{n=1}^{N_1} R_n}{t_{\mathrm{SI}}}$ and $E_3 = P_{\mathrm{c}} t_3$, respectively.

2.3. Problem formation

Let $\mathbf{t} = [t_1, \ldots, t_M]^T$ and $\mathbf{Q}_{\mathrm{u}} = [\boldsymbol{q}_{\mathrm{u}}(1), \ldots, \boldsymbol{q}_{\mathrm{u}}(N_1)]^T$ be the task dwell time vector and the UAV trajectory vector, respectively. Introducing weight coefficients w_1, w_2 and w_3, the considered problem is formulated as

$$\mathcal{P}: \max_{\mathbf{t}, t_{\mathrm{SI}}, \mathbf{Q}_{\mathrm{u}}} U(\mathbf{t}, t_{\mathrm{SI}}, \mathbf{Q}_{\mathrm{u}}) = w_1 p_{\mathrm{d}} + w_2 p_{\mathrm{g}} + w_3 R_{\mathrm{c}} \tag{10}$$

$$\text{s.t.} \quad \sum_{i=1}^{M} t_i \le t_{\mathrm{SI}} \le \frac{T}{N}, \forall i \tag{10a}$$

$$t_i^{\mathrm{s}} + t_i \le t_j^{\mathrm{s}} < t_j^{\mathrm{s}} + t_j, \forall i, j \in \mathcal{M}, i < j \tag{10b}$$

$$\sum_{i=1}^{M} (P_{\mathrm{h}} t_i + E_i) \le P_{\mathrm{av}} t_{\mathrm{SI}}, \forall i \tag{10c}$$

$$R_{\mathrm{c}} \ge R_{\mathrm{m}} \tag{10d}$$

$$\|\mathbf{q}_{\mathrm{u}}(n+1) - \mathbf{q}_{\mathrm{u}}(n)\|_2 \le \gamma v_{\max}, \forall n \tag{10e}$$

where $U(\mathbf{t}, t_{\mathrm{SI}}, \mathbf{Q}_{\mathrm{u}})$ denotes the joint performance; t_i^{s} denotes start time of task i; P_{h} is the power consumed by the UAV itself; P_{av} represents the average power consumed by the UAV while performing these three tasks; R_{m} is the minimum transmission rate of the UAV to perform the communication task; (10e) is the maximum velocity limit of the UAV.

3. Algorithm analysis

In this section, the considered non-convex problem $\mathcal{P}$ is solved by an AIO algorithm. For details, please refer to the analysis of the algorithm below.

3.1. Dwell time

When t_{SI} and $\mathbf{Q}_{\mathrm{u}}$ are fixed, $\mathcal{P}$ is rewritten as

$$\mathcal{P}1: \max_{\mathbf{t}} \quad U(\mathbf{t}) \tag{11}$$

$$\text{s.t.} \quad (10a)\sim(10d) \tag{11a}$$

Since (11) and (11a) are both convex, $\mathcal{P}1$ is considered as a convex optimization problem. Thus, the existing standard convex algorithms, such as interior point (IP) method [10], can be utilized to solve this problem.

3.2. Scheduling interval

When $\mathbf{t}$ and $\mathbf{Q}_{\mathrm{u}}$ are fixed, $\mathcal{P}$ is rewritten as

$$\mathcal{P}2: \max_{t_{\mathrm{SI}}} \quad U(t_{\mathrm{SI}}) \tag{12}$$

$$\text{s.t.} \quad (10a), (10c) \tag{12a}$$

Lemma 1. *$p_g(t_{\mathrm{SI}})$ is concave.*

Proof. Let $x = 1/t_{\mathrm{SI}}$. If t_{SI} is infinite and the other parameters in (7) are ignored, x is infinitely small, then

$$p_g(x) \simeq x^4 \int_{-L_{\mathrm{v}}}^{L_{\mathrm{v}}} \exp(-x^2)dz \int_{-L_{\mathrm{x}}}^{L_{\mathrm{x}}} \exp(-x^4+2x^3-x^2)dz \tag{13}$$

Assume that the area of the narrow strip rectangle is used as an approximation to the integral area of the surface, we have

$$\int_{-L_{\mathrm{v}}}^{L_{\mathrm{v}}} \exp(-x^2)dz \approx 2L_v \exp(-x^2) \tag{14}$$

$$\int_{-L_{\mathrm{x}}}^{L_{\mathrm{x}}} \exp(-x^4+2x^3-x^2)dz \approx 2L_x \exp(-x^4+2x^3-x^2) \tag{15}$$

Thus, $p_g(x) \approx x^4 \exp(-x^4+2x^3-2x^2)$. Define $f(x) = x^4 \exp(-x^4+2x^3-2x^2)$, and take the second-order derivation of $f(x)$ with respect to x,

$$\begin{aligned} f''(x) =& 4x^2[4x^6(x-3)^2 + 12x^5(x-1)^2 + 5x^2(x^2-1)^2 \\ &+ 3x^2(x+4)(x+1) - 26x^2 + 3] \\ &\times \exp(-x^4+2x^3-2x^2) \end{aligned} \tag{16}$$

According to the parameter setting in Section 4 and (10a), $t_{\mathrm{SI}} \geq 1$ and then $0 < x \leq 1$. After that, substituting $x = 1/t_{\mathrm{SI}}$ into Eq. (16), $f''(x) > 0$. Therefore, $p_g(x)$ is proved to be concave. Since $p_g(x)$ is an approximate function of $p_g(t_{\mathrm{SI}})$, then $p_g(t_{\mathrm{SI}})$ is also a concave function.

Thus, the proof is completed. ■

Since p_g is concave and R_{c} is convex, (12) is regarded as non-convex. To solve $\mathcal{P}2$, we convert (12) to a convex function. For ease of calculation, (12) is written as

$$U(t_{\mathrm{SI}}) = F(t_{\mathrm{SI}}) \tag{17}$$

where $F(t_{\mathrm{SI}}) = w_2 p_g(t_{\mathrm{SI}}) + w_3 R_{\mathrm{c}}(t_{\mathrm{SI}})$. Then, successive convex approximation (SCA) algorithm [11] is adopted to achieve a local optimal solution of $\mathcal{P}2$. Let t_{SI}^l be t_{SI} at the lth iteration, and the lower bound of $p_g(t_{\mathrm{SI}})$ is

$$\hat{F}(t_{\mathrm{SI}}, t_{\mathrm{SI}}^l) = F(t_{\mathrm{SI}}^l) + \nabla F(t_{\mathrm{SI}}^l)(t_{\mathrm{SI}} - t_{\mathrm{SI}}^l) \tag{18}$$

where $F(t_{\mathrm{SI}}^l)$ and $\nabla F(t_{\mathrm{SI}}^l)$ are $F(t_{\mathrm{SI}})$ at the lth iteration and the first-order derivation of $F(t_{\mathrm{SI}}^l)$ with respect to t_{SI}^l, respectively.

As a result, $\mathcal{P}2$ is rewritten as

$$\mathcal{P}21: \max_{t_{\mathrm{SI}}} \quad U'(t_{\mathrm{SI}}) = \hat{F}(t_{\mathrm{SI}}, t_{\mathrm{SI}}^l) \tag{19}$$

$$\text{s.t.} \quad (10a), (10c) \tag{19a}$$

$$\hat{F}(t_{\mathrm{SI}}, t_{\mathrm{SI}}^l) \geq F(t_{\mathrm{SI}}) \tag{19b}$$

With convex (19) and (19a)∼(19b), $\mathcal{P}21$ is convex. Furthermore, we can adopt the standard convex optimization techniques, such as CVX solver [11], to iteratively solve this problem. It is noted that the sub-optimal solution obtained from $\mathcal{P}21$ is a lower bound of $\mathcal{P}2$.

3.3. UAV trajectory

When $\mathbf{t}$ and t_{SI} are fixed, $\mathcal{P}$ is rewritten as

$$\mathcal{P}3: \max_{\mathbf{Q}_{\mathrm{u}}} \quad U(\mathbf{Q}_{\mathrm{u}}) \tag{20}$$

$$\text{s.t.} \quad (10d), (10e) \tag{20a}$$

Since (20) and (10a) are non-convex, $\mathcal{P}3$ is non-convex. At the kth iteration, the transmission rate of the UAV is rewritten as

$$R_n^k = \gamma B \log_2\left(1 + \frac{g_0}{H^2 + \|\mathbf{q}_{\mathrm{u}}^k(n) - \mathbf{Z}_0\|_2^2}\right) \tag{21}$$

where $g_0 = \frac{P_{\mathrm{c}}\rho_0}{n_0 B}$. Similarly, we also adopt to SCA algorithm, the first-order derivation of R_n^k with respect to $\|\mathbf{q}_{\mathrm{u}}(n) - \mathbf{Z}_0\|_2^2$ is

$$\hat{R}_n^k = \frac{\gamma B g_0}{H^2 + \|\mathbf{q}_{\mathrm{u}}^k(n) - \mathbf{Z}_0\|_2^2} \cdot \frac{\log_2(e)}{H^2 + \|\mathbf{q}_{\mathrm{u}}^k(n) - \mathbf{Z}_0\|_2^2 + g_0} \tag{22}$$

Then, the lower bound is

$$\hat{R}_n^k \geq R_n^k + \hat{R}_n^k \cdot (\|\mathbf{q}_{\mathrm{u}}(n) - \mathbf{Z}_0\|_2^2 - \|\mathbf{q}_{\mathrm{u}}^k(n) - \mathbf{Z}_0\|_2^2) \tag{23}$$

Define the lower bound of R_n as

$$R_n^{k,\mathrm{lb}} \triangleq R_n^k + \hat{R}_n^k \cdot (\|\mathbf{q}_{\mathrm{u}}(n) - \mathbf{Z}_0\|_2^2 - \|\mathbf{q}_{\mathrm{u}}^k(n) - \mathbf{Z}_0\|_2^2) \tag{24}$$

, and $R_{\mathrm{c}}^{k,\mathrm{lb}} = \frac{\sum_{n=1}^{N_1} R_n^{k,\mathrm{lb}}}{t_{\mathrm{SI}}}$.

Then, $\mathcal{P}3$ is converted to

$$\mathcal{P}31: \max_{\mathbf{Q}_{\mathrm{u}}} \quad U(\mathbf{Q}_{\mathrm{u}}) \tag{25}$$

$$\text{s.t.} \quad (10d), (10e) \tag{25a}$$

$$R_{\mathrm{c}}^{k,\mathrm{lb}} \geq R_{\mathrm{m}} \tag{25b}$$

With convex (25) and (25a)∼(25b), $\mathcal{P}31$ is convex. Following this, a similar method is used as in the previous subsection for solving $\mathcal{P}31$. Thus, joint optimization of dwell time, scheduling interval, and UAV trajectory is detailed in Algorithm 1.

Algorithm 1 Alternately Iterative Optimization Algorithm

1: Initialize the iterative index $r = 0$ and $\{\mathbf{t}^0, t_{\mathrm{SI}}^0, \mathbf{Q}_{\mathrm{u}}^0\}$;
2: **Repeat**
3: Solve $\mathcal{P}1$ to obtain $\boldsymbol{t}^{r+1}$ by using IP method for the given $\{t_{\mathrm{SI}}^r, \mathbf{Q}_{\mathrm{u}}^r\}$;
4: Solve $\mathcal{P}21$ by using SCA algorithm and CVX solver to obtain t_{SI}^{r+1} for the given $\{\mathbf{t}^{r+1}, \mathbf{Q}_{\mathrm{u}}^r\}$;
5: Solve $\mathcal{P}31$ by using SCA algorithm and CVX solver to obtain $\mathbf{Q}_{\mathrm{u}}^{r+1}$ for the given $\{\mathbf{t}^{r+1}, t_{\mathrm{SI}}^{r+1}\}$;
6: Update the iterative index $r = r + 1$;
7: **Until** $r \geq N_{\max}$ or
8: $|U(\mathbf{t}^{r+1}, t_{\mathrm{SI}}^{r+1}, \mathbf{Q}_{\mathrm{u}}^{r+1}) - U(\mathbf{t}^r, t_{\mathrm{SI}}^r, \mathbf{Q}_{\mathrm{u}}^r)| \leq \delta$;
9: Output:Optimal U^*.

3.4. Convergence

Lemma 2. *Algorithm 1 is guaranteed to converge to the suboptimal solution of $\mathcal{P}$.*

Proof. At the $(r+1)$th iteration, given $\{t_{\mathrm{SI}}^r, \mathbf{Q}_{\mathrm{u}}^r\}$, $\mathbf{t}^{r+1}$ can be obtained by solving $\mathcal{P}1$, and then we have

$$U(\mathbf{t}^{r+1}, t_{\mathrm{SI}}^r, \mathbf{Q}_{\mathrm{u}}^r) \geq U(\mathbf{t}^r, t_{\mathrm{SI}}^r, \mathbf{Q}_{\mathrm{u}}^r) \tag{26}$$

Next, given $\{\mathbf{t}^{r+1}, \mathbf{Q}_{\mathrm{u}}^r\}$, t_{SI}^{r+1} can be obtained by solving $\mathcal{P}21$, then we have

$$U(\mathbf{t}^{r+1}, t_{\mathrm{SI}}^{r+1}, \mathbf{Q}_{\mathrm{u}}^r) \geq U(\mathbf{t}^{r+!}, t_{\mathrm{SI}}^r, \mathbf{Q}_{\mathrm{u}}^r) \tag{27}$$

Finally, when solving the UAV trajectory sub-problem, given $\{\mathbf{t}^{r+1}, t_{\mathrm{SI}}^{r+1}\}$, $\mathbf{Q}_{\mathrm{u}}^{r+1}$ can be obtained by solving $\mathcal{P}31$, then we have

$$U(\mathbf{t}^{r+1}, t_{\mathrm{SI}}^{r+1}, \mathbf{Q}_{\mathrm{u}}^{r+1}) \geq U(\mathbf{t}^{r+1}, t_{\mathrm{SI}}^{r+1}, \mathbf{Q}_{\mathrm{u}}^r) \tag{28}$$

As a result, we obtain at the $(r+1)$th iteration

$$U(\mathbf{t}^{r+1}, t_{\mathrm{SI}}^{r+1}, \mathbf{Q}_{\mathrm{u}}^{r+1}) \geq U(\mathbf{t}^r, t_{\mathrm{SI}}^r, \mathbf{Q}_{\mathrm{u}}^r) \tag{29}$$

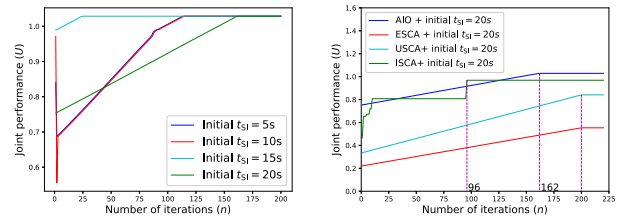
Thus, the proof is completed. ■

4. Simulation analysis

We consider a UAV-assisted RC system, where the UAV and the target at a fixed altitude $H = 100$ m as well as the communication node are randomly distributed within a two-dimensional area of 100×100 m^2. For detection task, we set $t_1 \in [1, 5]$ s and $p_{\mathrm{fa}} = 0.0015$, respectively. For tracking task, we set $t_2 \in [2, 10]$ s, $\sigma_{\mathrm{v}}^2 = 1.0$, $v = 5$ m/s, $L_{\mathrm{v}} = 100$ m/s and $L_{\mathrm{x}} = 16$ m, respectively. For communication task, we set $t_3 \in [1, 5]$ s, $R_{\mathrm{m}} = 5$ kb/s, $\rho_0 = -60$ dBm, $B = 10$ MHz and $n_0 = -110$ dBm/Hz, respectively. Other parameters are set to $T = 600$ s, $N = 30$, $N_1 = 5$, $P_{\mathrm{r}} = P_{\mathrm{av}} = 30$ W, $P_{\mathrm{h}} = P_{\mathrm{c}} = 10$ W and $\delta = 10^{-5}$, respectively.

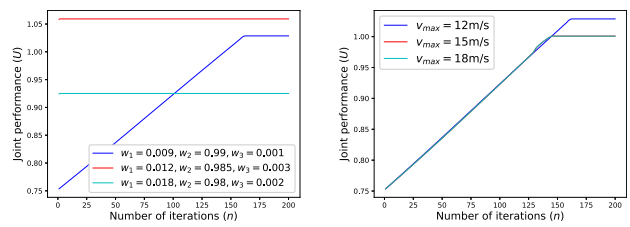
Then, we conduct simulation experiments for algorithm comparisons, i.e., the proposed AIO algorithm and three baselines: (1) Equal proportional task dwell time and SCA (ESCA); (2) Unequal proportional task dwell time and SCA (USCA); (3) Improved PSO [12] and SCA (ISCA).

Fig. 1 depicts an illustration of the suboptimal joint performance of four schemes applied to solving the considered problem. From Fig. 1(a), we can observe that as n increases, the joint performance of AIO with initial $t_{\mathrm{SI}} = 5$ s and $t_{\mathrm{SI}} = 10$ s first decrease fast and increase rapidly, while that of AIO with initial $t_{\mathrm{SI}} = 20$ s increases dramatically. Meanwhile, they keep plateauing at approximately $n \geq 120$. Moreover, the curve when initial $t_{\mathrm{SI}} = 15$ s converges rapidly and then remains steady at $n \geq 25$. The reason behind this phenomenon is that $t_{\mathrm{SI}} = 15$ s is very close to the suboptimal scheduling interval, while the other t_{SI} is far from that. And the curves with initial $t_{\mathrm{SI}} = 15$ s or $t_{\mathrm{SI}} = 15$ s find the local suboptimal solution first and then the global optimal solution. Besides, the optimal joint performance is both the same, i.e., 1.02, and the joint performance gaps between AIO with initial $t_{\mathrm{SI}} = 5$ s, $t_{\mathrm{SI}} = 10$ s and $t_{\mathrm{SI}} = 20$ s are narrow when these three curves increase. Thus, different initial t_{SI} only affects the number of iterations when using AIO. In Fig. 1(b), when initial $t_{\mathrm{SI}} = 20$ s, it can be seen that as n increases, the joint performance of ESCA and USCA increase linearly, while that of ISCA alternates between growing rapidly and remaining flat because the improved PSO algorithm is prone to generate local suboptimal solution. Then that of ESCA or USCA and ISCA remain stable when $n \geq 200$ and $n \geq 96$, respectively. Compared to ISCA, the joint performance of AIO is higher or lower with the increment of n, and then also remains stable at $n \geq 162$. In addition, we note that the suboptimal joint performance of these four schemes is not the same, namely 1.02, 0.97, 0.84, and 0.55, respectively. Meanwhile, the suboptimal joint performance of ESCA and USCA is obviously lower than that of AIO, which is close to that of ISCA, and the suboptimal joint performance gap between AIO and ISCA is narrow. The reason is that the suboptimal solutions of ESCA or USCA and ISCA are far from and close to that of AIO, respectively.



(a) (b)

Fig. 1. (a) Comparisons of AIO with different initial t_{SI}; (b) Comparisons of AIO and three baselines in terms of joint performance when initial $t_{\mathrm{SI}} = 20$ s.



(a) (b)

Fig. 2. Comparisons of AIO with different parameters when initial $t_{\mathrm{SI}} = 20$ s.

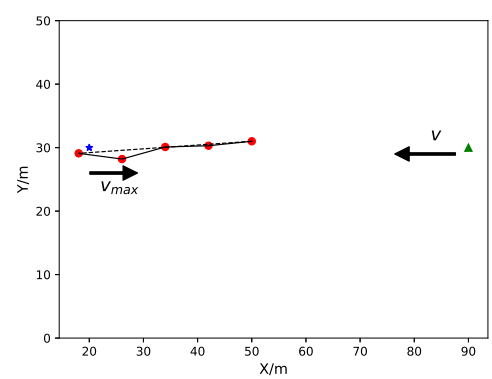


Fig. 3. UAV trajectory in two-dimensional plane. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

As shown in Fig. 2, we present AIO based on different parameters with respect to the number of iterations. From Fig. 2(a), we observe that the maximum values of the three curves hover around 1, and their values are 1.06, 1.02, and 0.96, respectively, in order. In addition, it is noted that two of the curves converge quickly, while the other one converges after several iterations, as in the case of the curve with $w_1 = 0.009, w_2 = 0.99, w_3 = 0.001$. In Fig. 2(b), the curve with $v_{\max} = 12$ m/s has the highest joint performance, roughly 1.02, while the other two curves have the same trend and joint performance of about 1. Therefore, we can observe from Fig. 2 that the effect of parameters on the maximum joint performance of the proposed algorithm is small.

Fig. 3 depicts the trajectory diagram of the UAV during a scheduling interval when initial $t_{\mathrm{SI}} = 20$ s, $v_{\max} = 12$ m/s, and $w_1 = 0.009, w_2 = 0.99, w_3 = 0.001$. In Fig. 3, we observe that the solid line between

the red dots indicates the trajectory of the UAV during the tasks, while the dashed line represents the return of the UAV to its initial position after completing the tasks. Besides, the green triangle and the blue pentagram indicate the target and the communication node, respectively.

5. Conclusion

In this paper, a joint detection, tracking, and communication optimization problem in a UAV-assisted RC system by jointly optimizing dwell time, scheduling interval, and UAV trajectory was described and formulated. After that, we proposed an AIO algorithm to solve the considered problem. Simulation results verified that the proposed algorithm can obtain higher joint performance than baseline schemes while reaching convergence. In addition, multiple dual-function UAVs to perform multi-target tasks will be studied in our future work.

CRediT authorship contribution statement

Xun Zhang: Writing – original draft, Software, Methodology, Conceptualization. **Xiaobai Li:** Writing – review & editing, Software. **Wei Cheng:** Writing – review & editing, Supervision. **Lu Gao:** Writing – review & editing, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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